

High Capacity Coherent Systems: EEPN Characterization, System Impact and Mitigation Options

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Abstract *Equalization-enhanced phase noise (EEPN) is a critical impairment in high-symbol-rate coherent optical systems. This paper analyses EEPN characteristics, system-level effects, and mitigation options. The dominant role of EEPN-induced timing jitter is demonstrated, and implications for pluggable transceivers across DCI, metro, and long-haul applications are discussed. ©2026 The Author(s)*

Introduction

The massive expansion of AI data centers requires high-capacity optical interconnections between distributed sites. For this purpose, coherent pluggable transceivers [1, 2] are increasingly deployed due to their technical suitability and cost advantages. As a result, pluggables now dominate applications ranging from DCI to metro networks. Demonstrations show their suitability even for much longer transmission distances [3].

The implementation of pluggable transceivers, however, involves several trade-offs to meet stringent requirements with respect to formfactor, power consumption, and cost. In high-symbol-rate systems, the linewidth (LW) of the local oscillator (LO) plays a particularly critical role. If the LW is too large, the phase noise interacts with the digital chromatic dispersion (CD) compensation, resulting in a distortion known as equalization-enhanced phase noise (EEPN) [4]. This effect scales with the symbol rate (SR), which has been continuously increased. Digital subcarrier multiplexing can reduce EEPN but introduces additional impairments [6]. As a result, EEPN sensitivity remains comparable in next-generation systems employing few subcarriers [7].

In this paper we discuss the fundamental characteristics of EEPN and system implications based on experiments, simulations and recently published papers. We compare the characteristics of EEPN across different transmission distances and application scenarios, including DCI, metro, long-haul, and even submarine systems, and conclude by discussing the challenges associated with EEPN compensation.

EEPN characteristics

With the introduction of coherent optical transmission systems approximately 20 years ago, CD compensation was shifted to the digital domain. To enable this transition without performance penalties, the optical phase of the receiver LO must remain sufficiently stable over the effective memory length of the CD filter. Early studies derived the corresponding requirements on the LO and quantified the resulting performance

degradations [4].

For many years, EEPN-induced distortions were considered only in terms of their average statistical behaviour [8, 9], while statistical anomalies were largely neglected. In early generations of coherent systems operating at symbol rates from 28 GBaud up to the fifth generation at around 90 GBaud, laser phase noise was comparatively low, with LWs on the order of only a few tens of kHz. As a consequence, EEPN-induced impairments were largely negligible. With the further increase of symbol rates beyond 100 GBaud, however, EEPN has again become a major impairment mechanism.

At the same time, the transition from embedded transceivers to pluggable modules has significantly altered the requirements imposed on the LO. In addition to LW, constraints related to cost, power consumption, and integrability must now be taken into account. This has resulted in the use of LOs with substantially increased LWs, for example up to 500 kHz for 800-Gb/s ZR systems and up to 150 kHz for ZR+ systems have been proposed recently [1, 2].

In [10], we analysed the characteristics of EEPN in detail and highlighted its burst-like distortion behaviour. EEPN leads to impairments that can persist over time intervals spanning many symbols. In the worst case, a large number of consecutive symbols may be strongly distorted, potentially even by an approximately constant impairment. Such severely distorted symbols can be regarded as statistical outliers.

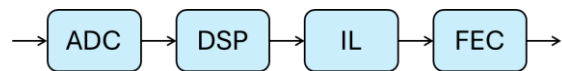


Fig. 1. Schematic architecture of the receiver.

To improve receiver sensitivity, modern coherent systems employ powerful forward error correction (FEC) schemes. These schemes rely on a favourable statistical distribution of outliers, as their capability to correct burst errors is inherently limited. To better distribute severely distorted symbols over time, interleavers (ILs) are used to spread the data across larger symbol blocks before decoding the data in the FEC (see Fig. 1). However, due to complexity constraints, the IL

depth cannot be increased arbitrarily. As a result, while the statistics of non-Gaussian impairments are significantly improved, they are not fully Gaussianized.

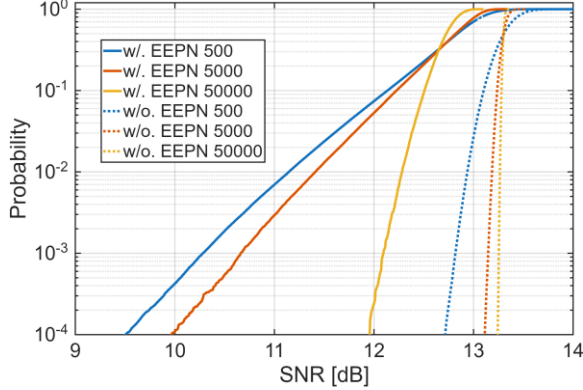


Fig. 2. CCDF of block-wise SNR for CD=54 ns/nm, LW=115 kHz und SR=130 GBaud.

To investigate the impact of the block length on the signal-to-noise ratio (SNR) distribution (SNR), we performed EEPN system simulations. Fig. 2 shows the complementary cumulative distribution function (CCDF) of the SNR for block lengths from 500 up to 50,000, both with and without EEPN. In the absence of EEPN, the CCDF exhibits a very steep slope, particularly for large block lengths. Shorter block length lead to slightly worse blocks even for Gaussian distributed data. The CCDF at low probabilities defines the available worst-case SNR, which must exceed the minimum SNR required for successful FEC decoding. Consequently, the steepness of the CCDF has a critical impact on the FEC threshold and the resulting sensitivity penalties in system operation. In presence of EEPN, the CCDF slope becomes significantly flatter. In particular this highlights the drastic deterioration in a few blocks, which significantly reduce the system's worst SNR and drastically lower the FEC threshold.

System Implications of EEPN

EEPN causes distortions which exhibit fundamentally different effects on the received signal [10-12]. Among these distortions, timing errors, commonly referred to as timing jitter, cause the dominant performance penalty. The EEPN-induced timing jitter scales with the square root of the symbol rate, the square root of the LO LW, and approximately with the square root of the accumulated CD [13]. Fig.3 shows the RMS jitter obtained from experiments performed with LOs featuring LWs ranging from 30 kHz to 500 kHz.

The RMS timing jitter shows a strong dependence on both the LO LW and the accumulated CD. E.g. for e.g. 500 kHz LW LO, a 300 fs RMS jitter is reached already at roughly 20 ns/nm. A 30 kHz LW LO lowers the RMS jitter drastically allowing to extend the CD range beyond

200 ns/nm at same 300 fs timing jitter.

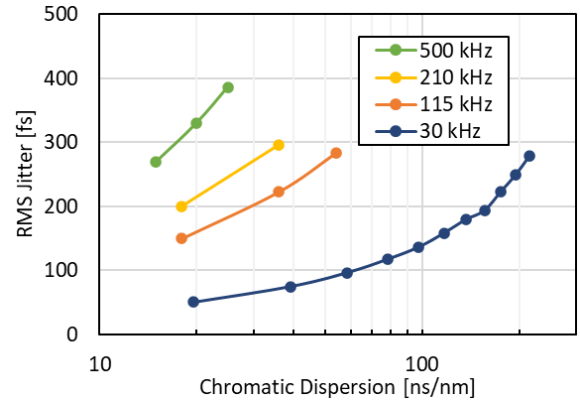


Fig. 3. RMS timing jitter vs. CD. Lasers with LWs from 30 to 500 kHz have been used. SR=130 GBaud.

The evolution of timing jitters power spectral density (PSD) with dispersion is shown in Fig. 4. For the highest dispersion of ~400 ns/nm, we have an almost constant PSD up to 1 MHz. Beyond that, we observed a drop of the PSD of roughly 3 decades in magnitude within a single decade in frequency. After that, the PSD levels converge into a high frequency plateau that depends only weakly on the accumulated dispersion. Lowering the dispersion drastically leads to a reduction of the low-frequency plateau. The PSD then converges from the low frequency plateau to the same decay as with a large CD; however, this increases the cutoff frequency of the low-frequency plateau later referred to as f_{3dB} . It is exactly this low-frequency plateau of the timing jitters PSD that is caused by EEPN. Earlier we reported an almost constant distortion over several symbols for an EEPN burst, which comes from the limited bandwidth.

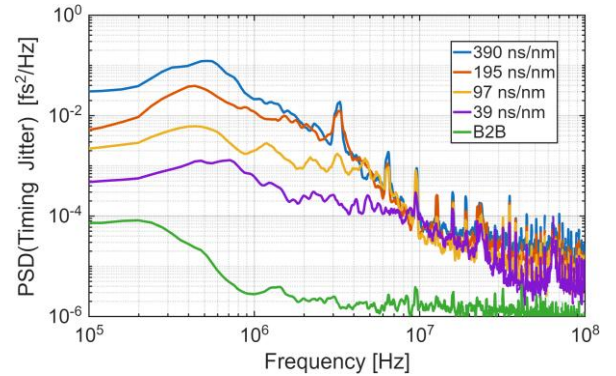


Fig. 4. PSD of timing jitter measured for an accumulated CD up to 390 ns/nm. SR=130 GBaud.

In [13] the bandwidth of the EEPN-induced timing jitter was approximated as

$$f_{3dB} \approx \frac{0.6}{2\pi\beta_2 l f_{sym}} \quad (1)$$

where β_2 is the group velocity dispersion, l the fiber length and f_{sym} the symbol rate frequency. According to (1) the jitter bandwidth decreases with same factor as the fiber length increases for a

constant symbol rate. Furthermore, it does not depend on the LO LW. For metro transmission distances of e.g. 1000 km, corresponding to an accumulated dispersion of about 20 ns/nm, the jitter bandwidth is in the range of 30 MHz [14].

Further contributions of EEPN are listed in [8] according to the frequency dependent phase walk-off or in [9] according to its effect onto the output signal. Overall, these components are far smaller than the timing jitter caused by EEPN, in [11] half penalties were reported for the second largest contribution.

Mitigation options

Effective mitigation of EEPN-induced distortions requires consideration of both the distortion characteristics themselves and their temporal dynamics. Depending on the phase noise properties, there exist shorter or longer time intervals during which EEPN primarily manifests as a frequency-dependent phase distortion [14–16]. This allows the degrees of freedom of the compensation filter to be reduced to just one: the frequency-dependent phase [16, 17].

In general, frequency-dependent phase distortions can be mitigated using linear filtering, most notably feed-forward equalizers (FFE). Several approaches have been reported in the literature, [18, 19], where remarkable performance gains have been demonstrated. However, practical FFE implementations typically rely on adaptive feedback loops, whose convergence speed is severely limited not only by the feedback structure itself, but also by the latency of the filter and the feedback path. As a result, such equalizers become comparatively slow, and the achievable performance gains are reduced in real systems.

As an alternative, timing-estimation (TE) approaches have been proposed in [20, 21]. One key advantage of these methods is their ability to directly suppress timing jitter using zero-forcing algorithms, in which the timing error of an entire data block is explicitly estimated and compensated. Furthermore, this architecture is less critical with respect to timing boundaries within the signal processor, since individual blocks only need to be buffered for a short duration before the correction is performed. No interaction with other processing stages is required, making these algorithms well suited for parallel implementations.

The effective speed of TE schemes is determined by the block size and the corresponding update frequency. In [14], we experimentally investigated the influence of the TE bandwidth on the timing jitter mitigation. For an accumulated dispersion of 18 ns/nm corresponding to 950 km transmission distance, the jitter bandwidth was found to be around 30 MHz, which is close to the maximum achievable update rate of the used TE algorithm. Under these conditions, a reduction of

the RMS timing jitter by 36% was achieved. At higher dispersion of 54 ns/nm the jitter mitigation was increased to 44%. While the improvement in average SNR was negligible, a substantial enhancement was observed for severely degraded data blocks (cf. Fig. 5). At a probability level of 10^{-4} , SNR gains exceeding 1 dB were obtained in the CCDF. Comparing the experimental results from [14] with simulations we found almost 50% reduction of the EEPN penalty by the fast TE.

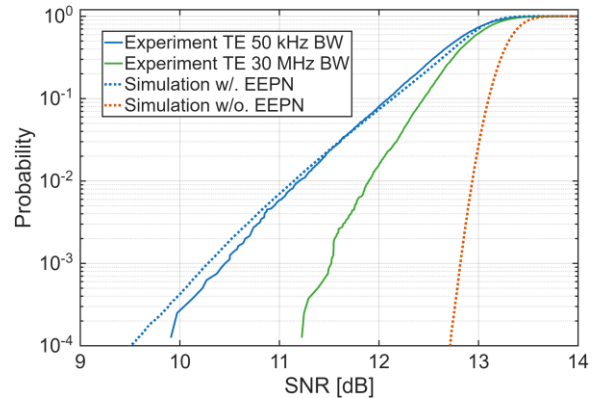


Fig. 5. CCDF of block wise SNR for experiments (solid) and simulations (dotted).

Conclusions

Equalization-enhanced phase noise is a critical impairment for both current and future generations of coherent optical transmission systems. Among the various EEPN-induced distortions, the timing jitter constitutes the dominant contribution. This timing jitter scales with the LO LW, the accumulated CD, and the SR. An additional key characteristic of the EEPN-induced timing jitter is its limited bandwidth, which for metro systems typically lies in the range of roughly 10...30 MHz.

This bandwidth limitation enables effective suppression of timing jitter using fast digital signal processing blocks, such as TE algorithms. Based on these findings, the timing requirements for EEPN mitigation are relaxed for longer transmission distances due to the reduced jitter bandwidth. In contrast, for shorter transmission distances the jitter bandwidth increases, rendering EEPN suppression increasingly challenging. For the shortest coherent transmission distances, in particular ZR-type applications, emphasis must therefore be placed on ensuring that the timing jitter remains sufficiently small for the intended use case.

In summary, a detailed understanding of the characteristics of EEPN-induced distortions allows optimal strategies for managing and mitigating this impairment to be derived. Such insight facilitates the development of appropriate equalization and compensation techniques and enables an extension of the applicable operating range of laser sources, particularly local oscillators, with respect to their phase noise characteristics.

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